

E.S.P.: Extra-Sensory Puck in Air Hockey using the Projection-Based Illusion

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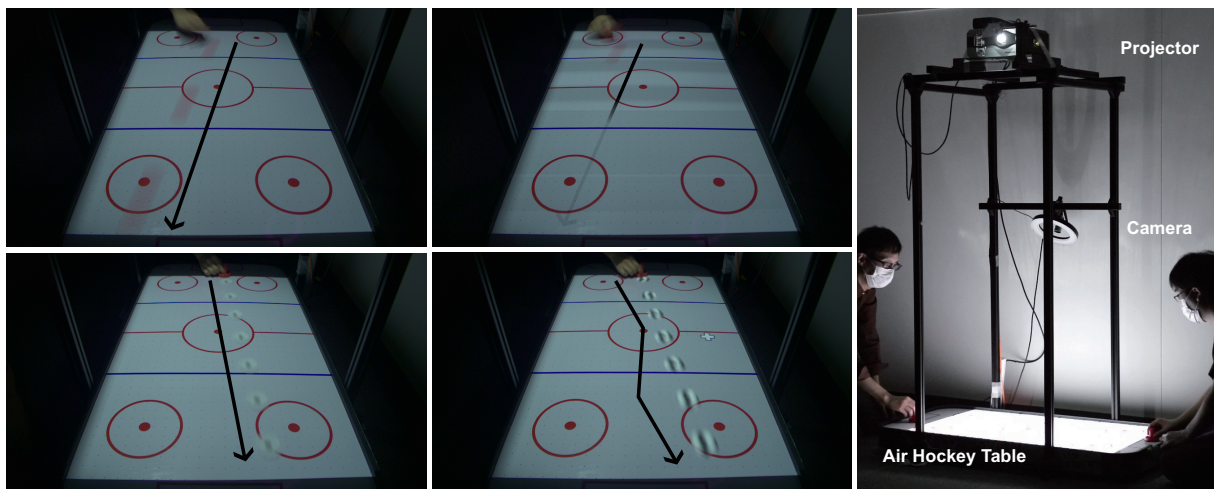


Figure 1: Invisible puck: Under our projection-based illusion, the perception of the moving puck (top left) is altered to be invisible (top center). **Altered motion:** The puck is heading in a straight line (bottom left), but its trajectory is perceived to be bent into an S-shape curve (bottom center). The right figure shows our system projecting the illusion-inducing patterns.

ABSTRACT

E.S.P. (Extra-Sensory Puck) provides a new experience by introducing optical illusions to air hockey. The perception of a solid puck randomly hit by a player is altered to show various physics-defying appearances and motions to the player's naked eye. Such altered perceptions are based on our high-speed projector-camera system generating stimulation patterns onto the moving puck. This paper presents two demonstrations. The first is the *invisible puck*, where the hit puck is camouflaged to disappear on the table. The second is the *altered motion*, where the direction and speed are altered.

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1 INTRODUCTION

Optical illusions allow us to view the world with altered perceptions of motions and other visual special effects. Recent studies show that a projector is crucial in creating such illusions in the real world that can be experienced with the naked eye. In existing studies, viewers perceive a static and rigid object as deformed [Kawabe et al. 2016], see a wobbly slow-motion effect of guitar strings [Fukushima and Naemura 2017], or see an animated image made of real wool [Asahina et al. 2021; Yoshida et al. 2016]. Although these studies were applied to a static object or an object with a known motion, more recent studies show that the deformation illusion can be applied to a freely and dynamically moving rigid object on the basis of a projection controlled by high-speed visual tracking [Fukamizu et al. 2020].

We explored a new experience by introducing this projection-based optical illusion to sports for amusement purposes. Sports

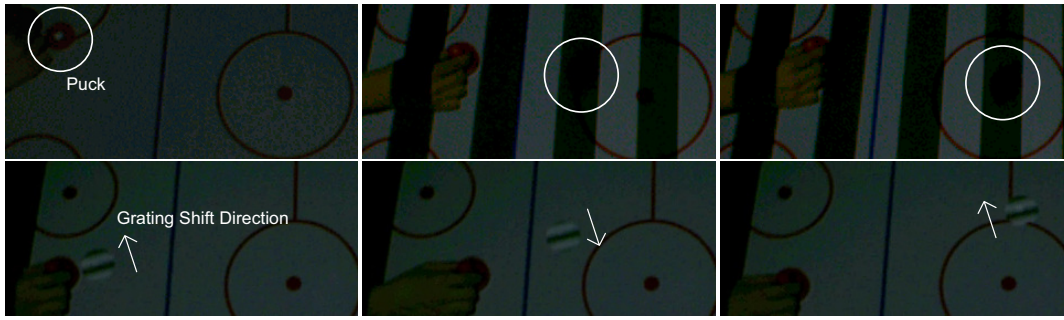


Figure 2: Projected patterns captured by a high-frame-rate camera. Top figures show the patterns for the *invisible puck* before and after hitting the puck. Bottom figures show the patterns for the *altered motion*.

present more challenging scenarios than the prior projection-based systems because more dynamic and interpersonal interactions occur. As a promising candidate sport, we focused on air hockey. Although several studies have reported augmentation of air hockey by projecting virtual mallets or pucks on the table [Sony Corporation 2018; Ueda et al. 2021], our proposed approach employs a different style called *E.S.P. (Extra-Sensory Puck)*. Our system can add illusions to a randomly hit physical puck to cause appearances or motions that do not obey physical laws.

2 SYSTEM

The system consists of a high-speed infrared (IR) camera, a high-speed monochrome projector [Watanabe et al. 2015], and the air hockey table, as shown in Figure 1. The IR camera obtains the puck’s location at 500 fps and the projector illuminates the puck or table with a stimulation pattern at 1000 fps, thereby inducing a visual illusion. The system compensates for the misalignment between the pattern and moving puck by predicting the puck motion. The misalignment is due to the latency from the motion to projection, which is 7.6 ms in this system.

3 DEMONSTRATIONS

3.1 Invisible puck

In this demonstration, a moving real puck appears invisible to the viewer. We leverage the flicker fusion effect for this illusion. As shown at the top of Figure 2, we project a moving stripe pattern at a constant speed. The speed and width of the black and white areas of the stripe pattern are adjusted after the puck is hit such that the puck remains within the black area in the pattern during its motion. Any point on the table is alternately illuminated by the white areas of the stripe pattern and darkened by the black areas of the stripe pattern as the stripe pattern moves across the table at the same temporal frequency, causing a “blinking” effect. The blinking frequency exceeds the flicker fusion threshold if the puck motion is fast enough. For example, if the threshold is 30 Hz and the puck diameter is 50 mm, the speed must be higher than 3.2 m/s, which is slow enough for the human eye to track under normal illumination. As a result of flicker fusion, a viewer does not notice the projected pattern. The viewer perceives a uniformly illuminated table where only the moving puck visually disappears, as shown in Figure 1.

3.2 Altered motion

In this demonstration, the direction and speed of the moving puck appears to be altered. We leverage the curve ball illusion that was developed for vision science research and tested only on monitors [Shapiro et al. 2010]. When altering the perceived direction, we project onto the moving puck a sinusoidal grating that shifts in a direction perpendicular to the direction of the actual puck motion, as shown in Figure 2. When the puck is observed with peripheral vision, the viewer perceives a modified motion trajectory. Figure 1 shows an unrealistic S-curved motion created by changing the grating shift direction twice¹. The projected cross shape in Figure 1 is the guide for the fixation point to see the puck with peripheral vision. In a similar manner, we can change the apparent speed of the puck motion by projecting the grating shifted in a direction parallel to the direction of the actual puck and changing the shifting speed during the puck motion.

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REFERENCES

- Ray Asahina, Takashi Nomoto, Takatoshi Yoshida, and Yoshihiro Watanabe. 2021. Realistic 3D Swept-Volume Display with Hidden-Surface Removal Using Physical Materials. In *IEEE VR*. 113–121.
- Kentaro Fukamizu, Leo Miyashita, and Masatoshi Ishikawa. 2020. ElaMorph Projection: Deformation of 3D Shape by Dynamic Projection Mapping. In *ISMAR*. 164–173.
- Shogo Fukushima and Takeshi Naemura. 2017. Wobble Strings: Spatially divided stroboscopic effect for augmenting wobbly motion of string instruments. *Entertainment Computing* 19 (2017), 101–111.
- Takahiro Kawabe, Taiki Fukiage, Masataka Sawayama, and Shin'ya Nishida. 2016. Deformation Lamps: A Projection Technique to Make Static Objects Perceptually Dynamic. *ACM Trans. Appl. Percept.* 13, 2 (2016), 1–17.
- Arthur Shapiro, Zhong-Lin Lu, Chang-Bing Huang, Emily Knight, and Robert Ennis. 2010. Transitions between Central and Peripheral Vision Create Spatial/Temporal Distortions: A Hypothesis Concerning the Perceived Break of the Curveball. *PLOS ONE* 5 (2010), 1–7.
- Sony Corporation. 2018. *A(i)R Hockey*. Retrieved September 10, 2022 from <https://www.sony.com/en/SonyInfo/design/stories/AiRHockey/>
- Shun Ueda, Shingo Kagami, and Koichi Hashimoto. 2021. MetamorHockey: A Projection-based Virtual Air Hockey Platform Featuring Transformable Mallet Shapes. *ACM SIGGRAPH 2021 Emerging Technologies*.
- Yoshihiro Watanabe, Gaku Narita, Sho Tatsuno, Takeshi Yuasa, Kiwamu Sumino, and Masatoshi Ishikawa. 2015. High-speed 8-bit Image Projector at 1,000 fps with 3 ms Delay. In *IDW*. 1064–1065.
- Takatoshi Yoshida, Yoshihiro Watanabe, and Masatoshi Ishikawa. 2016. Phyxel: Realistic Display of Shape and Appearance Using Physical Objects with High-Speed Pixelated Lighting. In *UIST*. 453–460.

¹Please watch the supplementary video to check the results.